

Project Report
Int 217
Project
LOVELY PROFESSIONAL UNIVERSITY
PHAGWARA, PUNJAB



AIR Quality Analysis

SUBMITTED BY: Rishabh Kumar

Registration Number: 12300726

Section – K23GW

Roll no- 02

DECLARATION

I, Rishabh Kumar, hereby declare that the work done by me on “Excel Project: Air Quality Analysis” is a record of original work for the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science - Data Science, Lovely Professional University, Phagwara.

Signature

Name: Rishabh Kumar

Reg: No: 12300726

Signature

Mrs. Baljinder Kaur

UID: 27952

ACKNOWLEDGMENT

First and foremost, I would like to express my deepest gratitude to my college for providing me with the opportunity and resources to undertake this project.

I extend my sincere thanks to my Teacher, **Mrs. Baljinder Kaur**, for his invaluable guidance, constructive feedback, and constant encouragement throughout the project. His expertise and support were instrumental in achieving the objectives of this work.

Thank you all

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1. Introduction: In today's data-driven environment, the ability to visualize and interpret data efficiently is crucial. This project focuses on building an interactive dashboard in **Microsoft Excel** using the **AirQuality** dataset. The dataset contains key environmental indicators such as levels of pollutants, temperature, and corresponding time and date values.

The goal of this dashboard is to provide a clear, concise, and interactive overview of air quality trends and patterns. By utilizing Excel's features such as pivot tables, slicers, charts, and conditional formatting, we aim to transform raw data into a user-friendly tool for monitoring and analyzing air quality over time. This can assist environmental researchers, policy makers, and the general public in making informed decisions based on real-time or historical air quality metrics.

2. Source of Dataset: <https://catalog.data.gov/dataset/air-quality>

3. Dataset Preprocessing:

➔ Before building the dashboard, several preprocessing steps were carried out:

- Handling Missing Values NA or NAN
Missing numerical values were replaced with the average value of the respective column.
- Data Cleaning
Removed unnecessary columns and standardized text formatting such as location names.
- Column Selection
Selected key columns relevant for analysis such as State, Location, Temperature, pH, DO, Conductivity, BOD, Nitrate, and Coliform levels.
- Data Type Fixing
Ensured numeric columns were formatted correctly for accurate aggregation.

4. Analysis on Dataset:

Objective 1: **Measure Count According to Name**

i) **General Description**

Analyze how many air quality measurements have been recorded for each **pollutant name**, helping to identify monitoring focus and data distribution across different pollutants.

ii) **Specific Requirements**

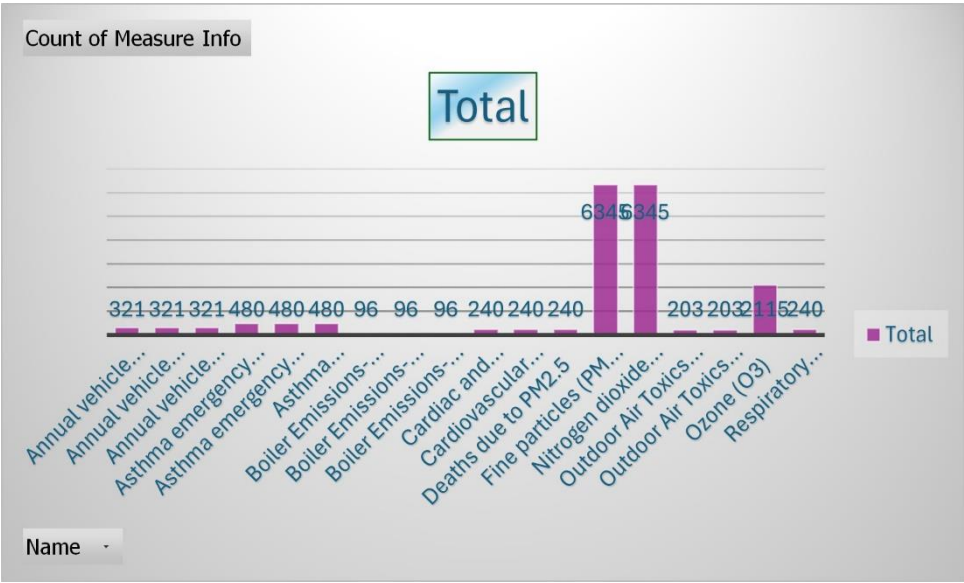
Create a **pivot table** with **Pollutant Name** (e.g., PM2.5, PM10, NO₂, O₃) in rows and **Count of Measurements** (records) in values.

iii) **Analysis Results**

Certain pollutants, such as **PM2.5** and **NO2**, have significantly higher measurement counts.

iv) **Visualization**

Use a **Vertical bar chart** to show the count of measurements per pollutant name, with **slicers** for **pollutant name** filtering.



Objective 2: Count of Measure Info

i) General Description

Assess how many air quality measurements have been recorded per geographic location to evaluate monitoring coverage and identify data concentration across regions.

ii) Specific Requirements

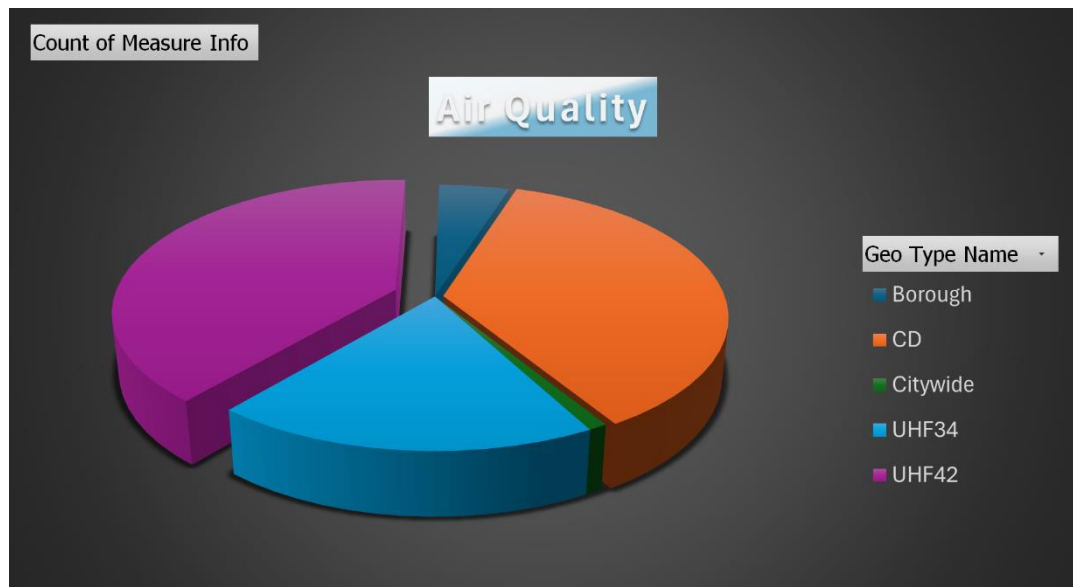
Generate a **pivot table** with **Geographic Place Name** in rows and **Count of Measurements** in values.

iii) Analysis Results

Some geographic areas show significantly higher counts, suggesting denser monitoring networks, while others have sparse data, indicating potential coverage gaps.

iv) Visualization

Use a **PieChart** showing measurement count by geographic area.



Objective 3: Count of Measurements

i) General Description

Track the total number of air quality measurements recorded to understand data volume, frequency, and trends in monitoring efforts.

ii) Specific Requirements

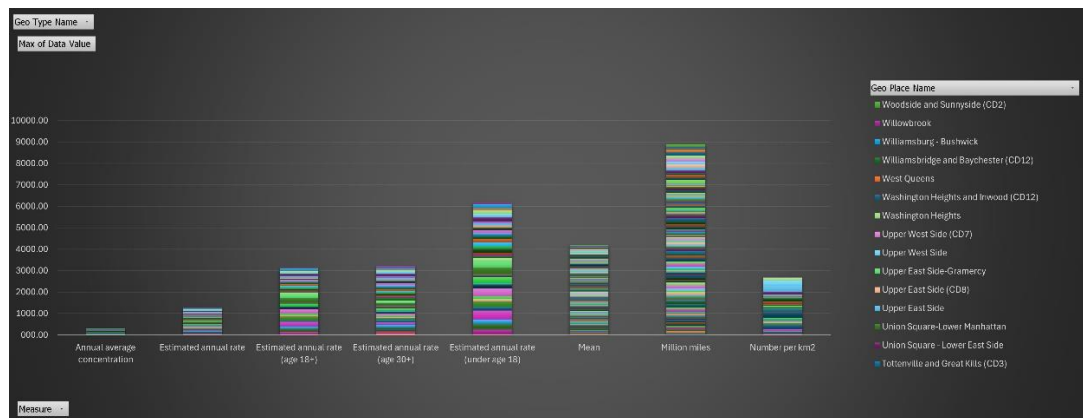
Create a **pivot table** showing the **total count of measurements**.

iii) Analysis Results

The overall count highlights fluctuations in data collection—some periods or stations show consistent monitoring, while others have gaps or lower activity.

iv) Visualization

Display a **Bar Graph** (for total count). with **slicers** for **Measure** to explore the data dynamically.



Objective 4: Air Pollution Monitoring Analysis

i) General Description

This analysis explores the most frequently monitored air pollutants in the dataset, focusing on how often different pollutants and measurement units appear. The goal is to identify key environmental indicators that are prioritized in urban air quality monitoring efforts.

ii) Specific Requirements

To determine:

- Which air pollutants are most frequently monitored.
- Which measurement units are most commonly used.
- How these pollutants are distributed across geographic zones.

iii) Analysis Results

1. Most Monitored Pollutants:

- Fine particles (PM2.5) – 6,285 records
- Nitrogen dioxide (NO2) – 6,234 records
- Ozone (O3) – 2,035 records

2. Common Measurement Units:

- mcg/m³ (micrograms per cubic meter)
- ppb (parts per billion)

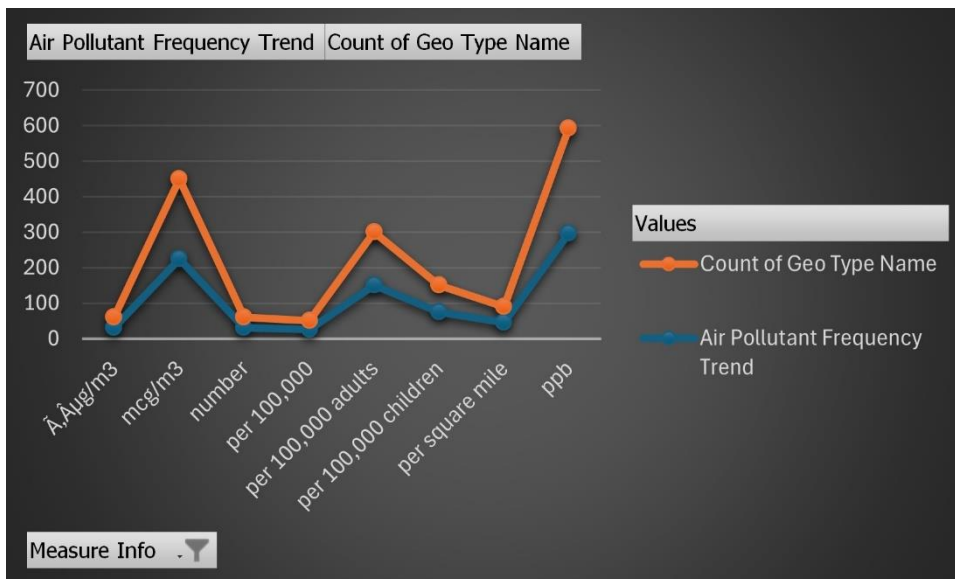
3. Geographic Coverage:

- Pollutants measured in mcg/m³ and ppb are consistently tracked across many geographic units (Community Districts, ZIP codes, etc.).

iv) Visualization

The dual-line chart above compares:

- Air Pollutant Frequency Trend (blue line)
- Count of Geo Type Name (orange line)



Objective 5: Sum of Data Values According to Geo Names

i) General Description

Calculate the total (sum) of recorded air quality data values for each geographic location to understand the overall pollutant load across regions.

ii) Specific Requirements

Create a pivot table with:

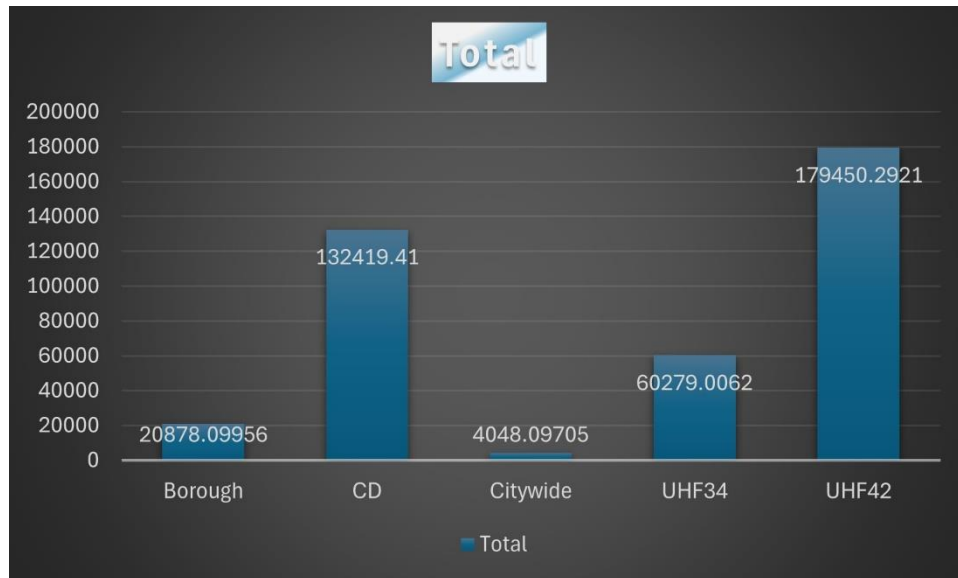
- Geo Place Name (e.g., City, District) in rows
- Sum of Data Values (e.g., PM2.5, PM10, NO₂, etc.) in values

iii) Analysis Results

Regions with higher cumulative pollutant values indicate areas with consistently high pollution levels or frequent monitoring. This helps identify pollution hotspots and prioritize areas for action.

iv) Visualization

- Clustered Bar Chart to compare total pollutant values across places



4. Conclusion:

The analysis highlights uneven air quality monitoring across locations, with some areas showing high pollutant levels and frequent measurements, while others lack data. Visual insights reveal pollution hotspots and data gaps, supporting informed decisions for targeted environmental action and improved air quality management.

LINKEDIN LINK:

https://www.linkedin.com/posts/rishabh-kumar-59b348299_airquality-dataanalysis-excel-activity-7317245035870048256_2KWW?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEghsUcBrzMfwlBdZb5ezCQFOsOqVQZs4M

5. Future Scope of Air Quality Analysis:

- Real-time monitoring and forecasting using IoT and machine learning
- Correlation of air quality with health outcomes
- Identification of localized pollution sources
- Evaluation of environmental policies and regulations
- Community-level awareness and engagement tools
- Integration with climate change and sustainability models

6. References:

Data Science Blog

Air Quality Data Analysis Using Python – Towards Data Science

URL: <https://towardsdatascience.com/air-quality-data-analysis>